Reg. No. | | | | | | | | | | | [12EC752]

SEVENTH SEMESTER B.E EXAMINATION DECEMBER - 2015

CAD FOR VLSI

Time:3 Hours **Maximum Marks:70**

Instructions: Answer any **FIVE** full Questions.

1. **a.** With neat diagram explain VLSI traditional design cycle. **6**

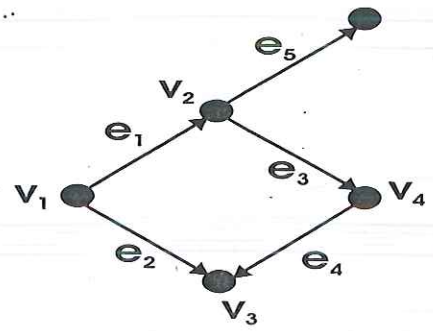
b. Explain the following VLSI design styles, with necessary diagrams; **8**

i. Full customs

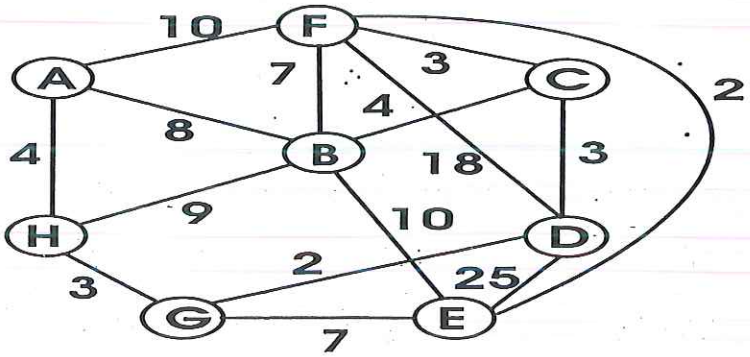
ii. Standard cell

iii. FPGA

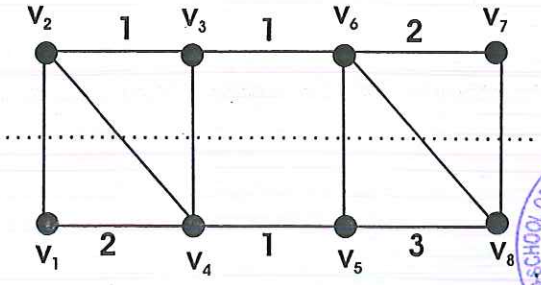
2. **a.** Explain depth first search algorithm for the graph shown in fig below: **4**



b. Explain the steps, that are involved in prim's algorithm to find minimum spanning tree (MST) for the graph shown **10**



3. Explain Kernighan–Lin partitioning algorithm for the graph shown in fig below **14**



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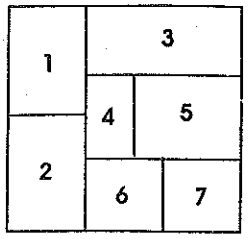
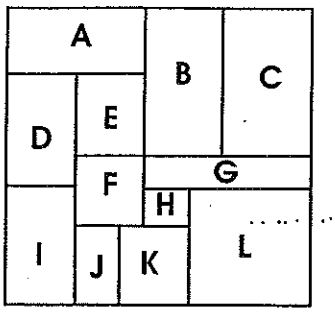
4. a. What is floor plan slicing? Draw skewed and non-skewed slicing tree for the floor plan shown in fig below, also give polish expression **6**



b. Draw the polar horizontal graph and polar vertical graph for the floor plan shown in fig below. **8**



5. a. Explain cluster- growth placement algorithm with a suitable example. **6**

b. Explain in brief, various wire length estimation methods that can be performed on a layout. **8**

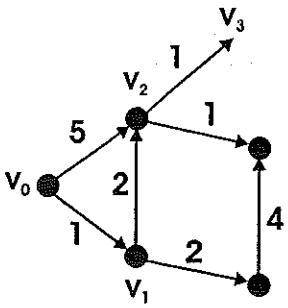
6. a. Explain maze-routing algorithm choosing a suitable example. **6**

b. With suitable example, discuss basic left- Edge algorithm (LEA) for two- layer channel routing **8**

7. a. Discuss the design rules which is scaled in λ. **3**

b. Draw symbolic and geometric layout of CMOS inverter. **5**

c. Explain longest path algorithm for compaction for the graph show in fig. **6**



8. a. How transistor is modeled by Bryant and the simulation is performed at switch level? **5**

b. How the signal, gate and delays are modelled for gate level simulation? **9**